Predicting Diabetes Risk from Behavioral CDC Data: A Comparative Analysis of Logistic Regression, Decision Trees, and K-Nearest Neighbors

Written by Alexandra Lostetter, Rohan Gogate, Rachel Xing, Yukai Li, Merry Li

Abstract

This project investigates the prediction of diabetes risk using behavioral survey data collected by the Centers for Disease Control and Prevention (CDC). The study focuses on comparing the performance of three machine learning models: Logistic Regression, Decision Trees, and K-Nearest Neighbors. Several strategies, including resampling techniques and feature engineering, are explored to address the class imbalance inherent in the dataset. Model effectiveness is evaluated using accuracy, precision, recall, specificity, and F1 score, with particular emphasis placed on recall due to the importance of correctly identifying individuals at risk for diabetes. The results provide insights into which modeling approaches are most effective when dealing with real-world health survey data.

1 Introduction

The early prediction of diabetes risk remains a crucial objective in public health, given the rising prevalence of the disease and its long-term complications. Behavioral survey data offers a valuable but challenging resource for prediction efforts, especially due to noise, missing values, and class imbalance. In this project, behavioral data from the CDC was utilized to build predictive models aimed at identifying individuals at heightened risk for developing diabetes.

The primary focus was a comparative analysis of three modeling approaches: Logistic Regression, Decision Trees, and K-Nearest Neighbors (KNN). Different resampling methods such as SMOTE, Random Oversampling, and NearMiss undersampling were applied to address the imbalance between diabetic and non-diabetic observations. Feature engineering efforts, such as the inclusion of interaction terms, were undertaken to enhance model performance. Models were evaluated using multiple performance metrics with a focus on recall, reflecting the importance of identifying as many at-risk individuals as possible.

2 Literature Review

In constructing predictive models for early diabetes detection, we benchmarked several machine learning algorithms based on their theoretical strengths, empirical performance, and practical operability with large imbalanced behavioral health datasets. Logistic Regression (LR) was selected as a baseline due to the fact that it possesses high interpretability, computational efficiency, and a strong statistical foundation, with easy odds ratio interpretations available to public health practitioners (1). However, LR has the drawback that it presumes a linear decision boundary, which may be violated with complicated real-world data (2), it is multicollinearity sensitive between predictors, and explicit missing data handling techniques are used. Decision Trees (DT) were also considered due to their ability to model non-linear relationships and interactions without requiring data normalization, and their rule-based nature, which in clinical environments makes them extremely interpretable (3). DTs, however, are susceptible to overfitting, creating very deep trees that memorize training data unless

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

regulated by pruning or depth constraints (4), are highly sensitive to minimal perturbations of the dataset, and can be biased towards dominant classes unless class imbalance is resolved (2). K-Nearest Neighbors (KNN), a flexible and non-parametric model that adapts to local structures in the data, was on the agenda due to its simplicity and structure interpretability (3) (6); however, KNN suffers from computational inefficiency whereby prediction time increases with the size of the growing training set, susceptibility to feature scaling wherein dominating features distort distance measures, and deteriorating performance in high-dimensional spaces because of the curse of dimensionality (2). Also, we considered Support Vector Machines (SVM) because of their very good theoretical generalization capability and ability to model complex, non-linear decision boundaries through the assistance of kernel functions. Yet SVM was also eliminated due to having serious practical disadvantages: it is computationally costly to the point of being infeasible on large data like the BRFSS survey (more than 250,000 observations) (7), it is very sensitive to class imbalance unless cost-sensitive adaptations are implemented, and it does not provide the interpretability necessary for clear communication in the public health field (1). Thus, although SVM is a tempting theoretical choice, issues of scalability, interpretability, and computational expense in the real world prompted us to focus our efforts on Logistic Regression, Decision Trees, and K-Nearest Neighbors.

3 Methods

3.1 Data Description

The dataset used in this study is derived from the 2015 Behavioral Risk Factor Surveillance System (BRFSS) survey, administered by the CDC and made available on Kaggle (7). The selected file, `diabetes_binary_health_indicators _BRFSS2015.csv`, contains 253,680 responses across 22 columns, reduced from the original 330 features based on relevance to chronic disease research. The target variable, `Diabetes_binary`, is categorized as 0 for no diabetes (84.7%) and 1 for prediabetes or diabetes (15.3%).

We further divide the predictor features into three types. Numerical (before encoding), including BMI (body mass index), MentHlth (poor mental health days), and PhysHlth (poor physical health days) Ordinal, including GenHlth (general health status), Age (13 age groups), Education (6 levels), Income (8 brackets), and Binary, which are Health and behavior indicators (0 = No, 1 = Yes), including HighBP, HighChol, CholCheck, Smoker, Stroke, HeartDiseaseorAttack, PhysActivity, Fruits, Veggies, HvyAlcoholConsump, AnyHealthcare, NoDocbcCost, DiffWalk, and Sex. To address class imbalance, we applied SMOTE (synthetic oversampling) and random oversampling, applied only to the training set to avoid data leakage. 24,206 duplicate rows were identified and removed.

Histograms and boxplots showed that BMI, MentHlth, and PhysHlth variables were right-skewed (Figure 1), with diabetic individuals exhibiting higher BMI and more days of poor health (Figure 2).Ordinal variable analysis indicated that poorer general health, older age, lower education, and lower income were associated with higher diabetes prevalence (Figure 3). Further analysis revealed higher diabetes rates among individuals with high blood pressure, high cholesterol, stroke, heart disease, physical inactivity, and difficulty walking.

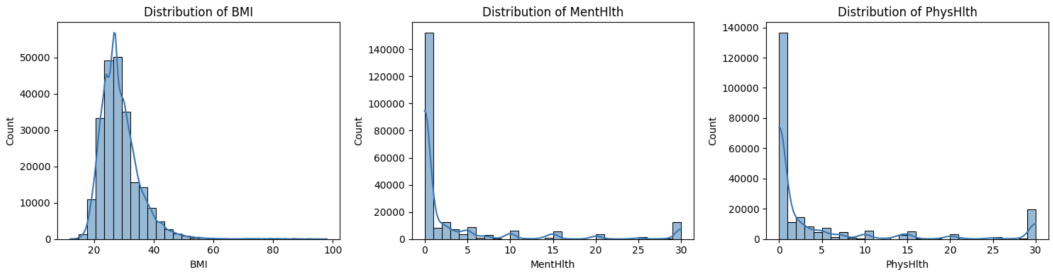

Figure 1: Distribution of BMI, mental health days, and physical health days

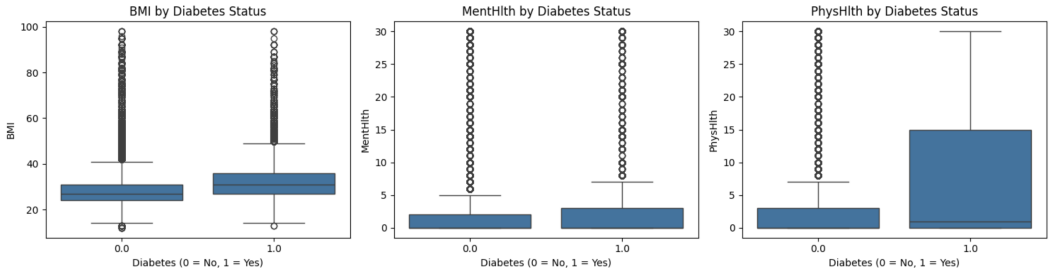

Figure 2: Boxplots of BMI, MentHlth, and PhysHlth by Diabetes Status

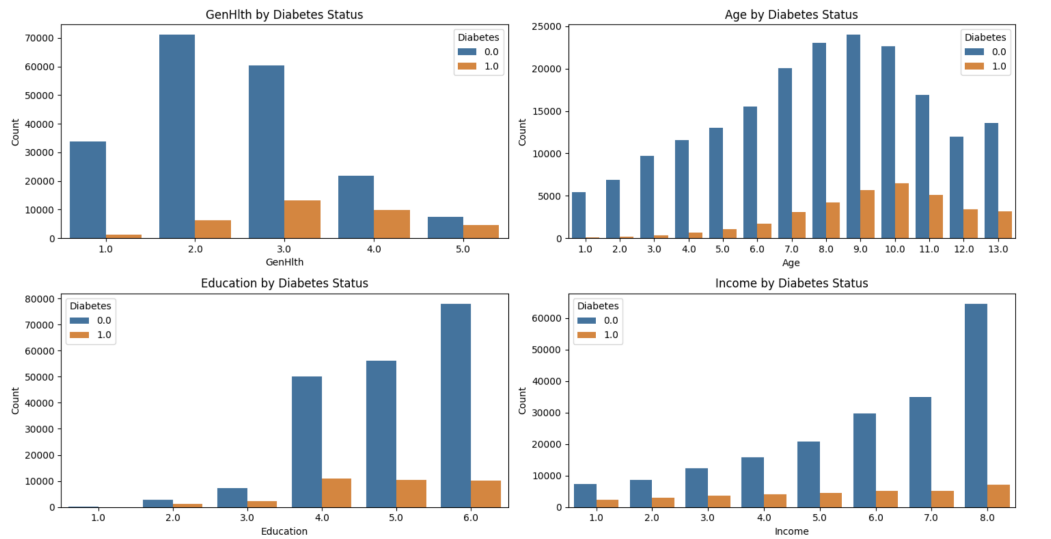

Figure 3: GenHlth, Age, Education, and Income Distribution by Diabetes Status

To explore relationships between variables and identify potential multicollinearity, we first calculated Pearson's correlation coefficients between each feature. Some initial findings revealed relatively high correlations, such as PhysHlth and GenHlth (0.52), GenHlth and DiffWalk (0.45), PhysHlth and DiffWalk (0.47), and Income and Education (0.42).

Recognizing the mixed data types, we applied appropriate correlation methods for different pairs: Point-Biserial for binary-numerical pairs, Pearson for numerical-numerical pairs, and Spearman for ordinal-binary and ordinal-numerical pairs. This analysis revealed moderate correlations, such as GenHlth and Income (-0.313), DiffWalk and Income (-0.285), DiffWalk and PhysHlth (0.467), and GenHlth and PhysHlth (0.439). To further assess multicollinearity, we calculated Variance Inflation Factors (VIF) for all numerical and ordinal variables, all of which were below 2, confirming that multicollinearity was not a concern. Therefore, no variables were removed.

3.2 Logistic Regression

This logistic regression analysis examines the model's performance on an imbalanced dataset by applying various resampling strategies and feature engineering techniques to improve accuracy and generalization. Given the large sample size, we employed the `saga` solver to support L1 and L2 regularization on large datasets.

The logistic regression estimates the probability of diabetes as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)}}$$

where y is the binary response variable, $x_1, \ldots, x_p$ are the predictor variables, and $\beta_0, \ldots, \beta_p$ are the model coefficients.

To prepare the features, we applied standard scaling to all numeric predictors to enhance optimization stability. Class imbalance was addressed by several ways. First we used the original unbalanced dataset and incorporate `class_weight='balanced'` to penalize the majority class more heavily. Additionally, we applied resampling techniques, including (1) SMOTE (Synthetic Minority Oversampling Technique), which synthesized new minority class samples by interpolating between existing minority class samples and their neighbors. This can be represented mathematically as:

$$\hat{x}_{new} = x_i + \lambda(x_j - x_i), \quad \lambda \in [0, 1]$$

(2) random oversampling, which duplicated existing minority class samples, and (3) NearMiss Undersampling, which selected majority samples closest to minority class instances.

Hyperparameters, including regularization strength $C \in \{0.01, 0.1, 1, 10\}$ and penalty type (L1 or L2), were optimized via a three-fold cross-validated grid search based on classification accuracy. For feature engineering, we introduced second-order interaction terms among top-ranked variable pairs identified by significance testing (p-values) from logistic regression analysis.

3.3 K-Nearest Neighbors (KNN)

We implemented a K-Nearest Neighbors (KNN) model to classify diabetes outcomes. It predicts the class of a new observation by identifying the K closest points in the training data, based on a distance metric such as Euclidean distance, and assigning the majority class among those neighbors. To find the optimal configuration, we performed hyperparameter tuning by varying the number of neighbors (K) across a range of 5 to 500 in steps of 10, using 2-fold cross-validation on a resampled training subset created with SMOTE. We also experimented with different KNN algorithms ('brute', 'kd_tree', and 'ball_tree'), and found that the 'brute' force method performed best for our data. This result aligns with expectations because our feature set contains many categorical and ordinal variables, making simple distance comparisons more effective than tree-based partitioning methods. To further assess the robustness of the KNN model, we conducted an additional experiment utilizing data resampled via RandomOverSampler instead of SMOTE. After applying random oversampling and subsequently downsampling 10% of the dataset to improve computational efficiency, we trained a simple KNN model with K = 35. Furthermore, the hyperparameter tuning visualizations reveal that the cross-validated accuracy initially increases and reaches a peak around K = 30–50, after which it gradually declines(Figure 4). Meanwhile, the cross-validated error rate follows an opposite trend, increasing steadily beyond the optimal K, which is consistent with the expected bias-variance tradeoff behavior in KNN models.

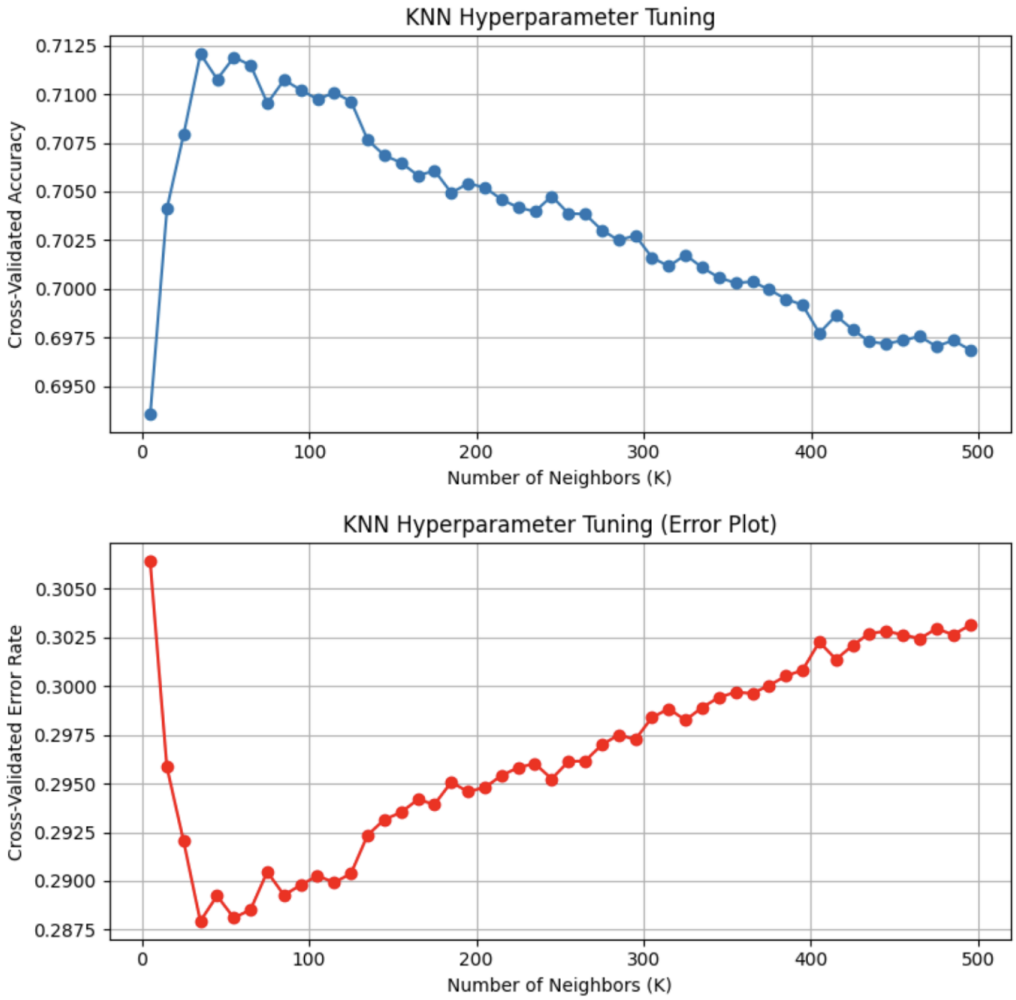

Figure 4: Tuning for optimal maximum depth value

3.4 Decision Trees

A Decision Tree Classifier model was developed to perform a binary classification task. It predicts the label of a new observation by following a series of binary splits based on feature values. To address the class imbalance in the original dataset, both SMOTE and simple oversampling techniques were utilized, and the results between these methods were compared. The model was optimized by determining the best depth value through plotting all possible values and their corresponding test scores to identify the "elbow point," where overfitting begins(Figures 5. This approach helped fine-tune the model for optimal performance. The result to this fine tuning that will be used later in the analysis was a depth of 2 for oversampling and 5 for SMOTE. SMOTE outperformed the oversampling approach, with testing accuracies of 0.801 and 0.742, respectively.

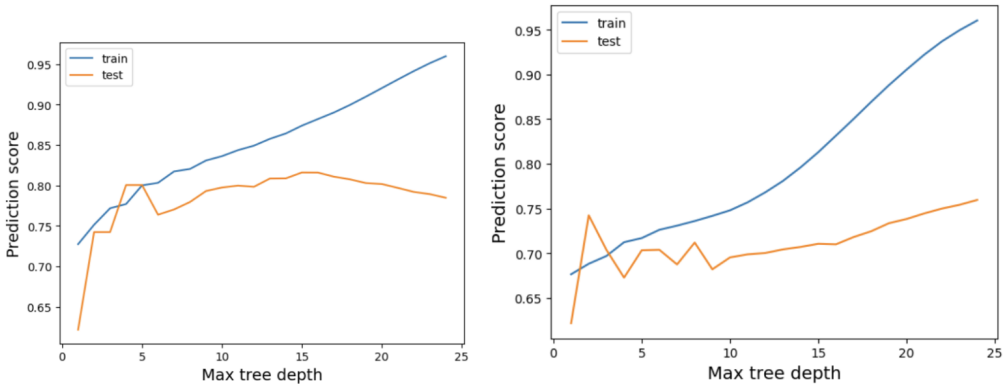

Figure 5: Tuning for optimal maximum depth value

4 Experiments and Results

4.1 Logistic Regression

Class weight balancing selected C=0.1 with an l2 penalty, achieving a training score of 0.717 and a test accuracy of 0.715. SMOTE selected C=10 with an l1 penalty, reaching a higher training score of 0.742, while random oversampling selected C=0.1 with an l2 penalty and a score of 0.733. All models achieved similar test accuracy around 71.5%, with SMOTE providing the strongest training performance.

To enhance predictive performance, we incorporated selected interaction terms into the logistic regression model based on statistical significance identified through an initial exploratory analysis. Specifically, interactions such as BMI × Age, Sex × Age, HighChol × Age, GenHlth × DiffWalk, and DiffWalk × Age were constructed and added to the original feature set. The inclusion of interaction terms can be modeled as:

$$P(y = 1 \mid x, z) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \gamma_1 (x_r z_s) + \cdots + \gamma_q (x_u z_v))}}$$

where: $x_1, \ldots, x_p$ are the original predictors, $(x_r z_s), (x_u z_v)$ are interaction terms (products of pairs of predictors), β are the coefficients for main effects, and γ are the coefficients for interaction terms.

The model achieved a training accuracy of 74.65%, a testing accuracy of 71.65%, and slightly increased recall, which is particularly important in imbalanced datasets.

NearMiss undersampling, coupled with feature scaling, produced the best results, with a training accuracy of 85.21% and a test accuracy of 84.93%. These findings suggest that undersampling the majority class can enhance generalization and overall model performance. Evaluation metrics like precision, recall, specificity, and F1 score were computed to evaluate model performance. Recall, as defined by:

$$\text{Recall} = \frac{TP}{TP + FN}$$

where TP is true positives and FN is false negatives, was particularly emphasized. The results indicate that SMOTE and the inclusion of interaction terms improved predictive performance, especially recall. These findings underscore the importance of selecting appropriate resampling techniques and feature engineering strategies when working with imbalanced datasets.

4.2 K Nearest Neighbors (KNN)

Hyperparameter tuning revealed that the optimal number of neighbors was K = 35 for both SMOTE and oversampling data, balancing training and validation performance. As K increased, we observed

that cross-validated accuracy first improved slightly, then gradually declined, reflecting the typical bias-variance tradeoff. The final KNN model trained on SMOTE-resampled data achieved a training accuracy of approximately 72.9% and a testing accuracy of 62.8%, with a validation error rate of 37.2%. Further evaluation based on confusion matrix analysis indicated a precision of 84.6%, recall of 62.7%, and an F1 score of 67.9%. These results suggest that while the model was highly precise in predicting positive diabetes cases, it struggled to capture all true positives, pointing to some limitations in sensitivity.

After applying RandomOverSampler, the model achieved a training accuracy of 73.4% and a testing accuracy of 67.4%, which, compared to the SMOTE-based model, demonstrates improved generalization ability, as indicated by the narrower gap between training and testing performance. The classification report on the test set shows a precision of 0.84, recall of 0.67, and F1-score of 0.72 (weighted averages). KNN demonstrated reasonable performance, but its reliance on distance metrics made it sensitive to the categorical-heavy feature space and data imbalance. While RandomOverSampler improved generalization compared to SMOTE, KNN still faced challenges in fully capturing minority class cases. (Figure6)

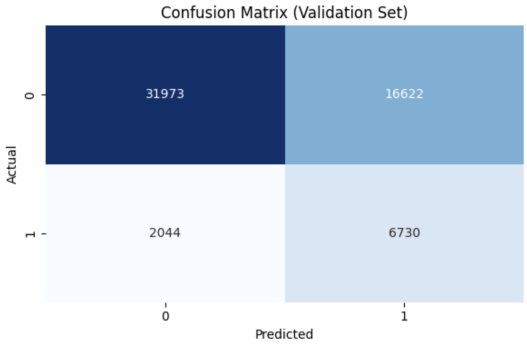

Figure 6: Confusion Matrix for KNN model

4.3 Decision Tree Performance

After fitting 5 for SMOTE and 2 for the oversampling dataset, the SMOTE and over sampled models achieved test set accuracies of 80% and 74%, respectively. However, further analysis indicated that the over sampled model was more appropriate for the medical context. Specifically, it had a lower False Negative Rate of 38% compared to SMOTE's 54%, which is critical, as missing a patient at risk for diabetes is far more costly than falsely flagging a healthy individual. Additionally, the over sampled model demonstrated a higher True Positive Rate, meaning it correctly identified a larger proportion of diabetics. A confusion matrix was used to analyze these results, and it was concluded that the SMOTE model underperformed due to the greedy nature of the CART algorithm(Figures 7). SMOTE generates synthetic samples by interpolating between existing ones, which may have introduced spurious patterns that the algorithm picked up, and these patterns did not generalize well.

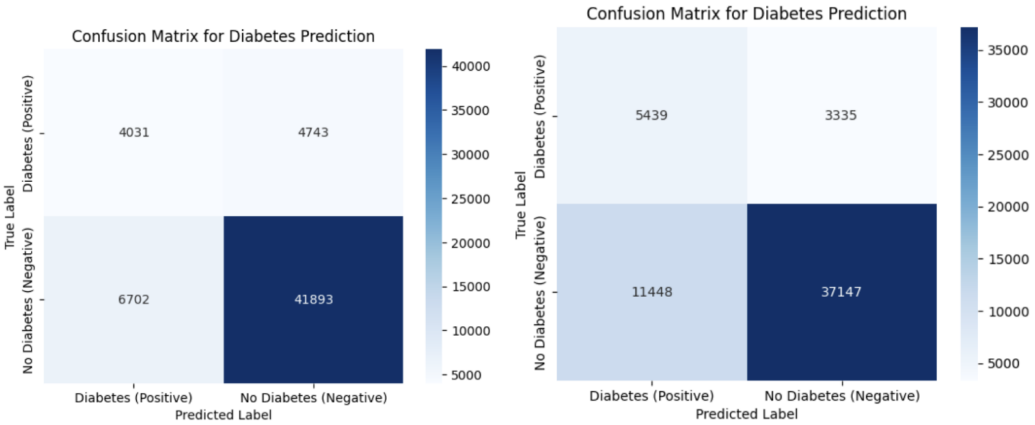

Figure 7: Confusion matrices for SMOTE and oversampled model respectively

Decision Tree models trained on oversampled and SMOTE data both highlight HighBP and GenHlth as the most important features. In the oversampled model, HighBP and GenHlth account for the majority of the predictive power. The SMOTE model similarly identifies HighBP and GenHlth as key features but also assigns notable importance to HighChol and Age as well, indicating a slightly broader set of influential variables.

5 Conclusion

This study highlights the challenges and potential solutions for predicting diabetes risk using behavioral survey data. Logistic Regression, Decision Trees, and K-Nearest Neighbors all have their strengths and weaknesses, with KNN and Decision Trees providing superior classification performance in this case. The results emphasize the importance of addressing class imbalance and applying proper resampling and feature engineering techniques to improve model effectiveness. Future work could explore ensemble methods like Random Forests and Gradient Boosting to further enhance prediction accuracy and robustness.

References

[1] P. Rajendra and S. Latifi. Prediction of diabetes using logistic regression and ensemble techniques. *Computer Methods and Programs in Biomedicine Update*, 1:100032, 2021. `https://doi.org/10.1016/j.cmpbup.2021.100032`

[2] N. Y. Philip, M. Razaak, J. Chang, S. M. M, M. O'Kane, and B. K. Pierscionek. A data analytics suite for exploratory, predictive, and visual analysis of type 2 diabetes. *IEEE Access*, 10:13460–13471, 2022. `https://doi.org/10.1109/ACCESS.2022.3146884`

[3] A. A. Al Jarullah. Decision tree discovery for the diagnosis of type II diabetes. In *2011 International Conference on Innovations in Information Technology*, pages 303–307. IEEE, 2011. `https://doi.org/10.1109/INNOVATIONS.2011.5893838`

[4] N. M. S. Kumar, T. Eswari, P. Sampath, and S. Lavanya. Predictive methodology for diabetic data analysis in big data. *Procedia Computer Science*, 50:203–208, 2015. `https://doi.org/10.1016/j.procs.2015.04.069`

[5] A. Ali, M. A. Alrubei, L. F. Hassan, M. A. Al-Ja'afari, and S. H. Abdulwahed. Diabetes diagnosis based on KNN. *IIUM Engineering Journal*, 21(1):175–181, 2020. `https://doi.org/10.31436/iiumej.v21i1.1206`

[6] S. Suriya and J. Joanish Muthu. Type 2 diabetes prediction using K-nearest neighbor algorithm. *Journal of Trends in Computer Science and Smart Technology*, 5(2):190–205, 2023. `https://doi.org/10.36548/jtcsst.2023.2.007`

[7] A. Teboul. Diabetes health indicators dataset. *Kaggle*, November 2021. `https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset`

Code Availability

The code implementation for this project is available at: https://colab.research.google.com/drive/1Xuoy_6d55YR7gvKGNOyHj17jnBFIfaTx#scrollTo=buAFcQLTwWys.